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COMPUTER PROGRAMMING PROJECT REPORT

GROUP 81

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ABSTRACT

This project focuses on the development of a python-based application for the analysis of statically determinate trusses using the method of joints. The application allows users to draw the truss on a grid, apply loads and define the supports in order to visualize the shape of the structure. The program was designed to make truss analysis very simple by automating calculations of member forces and joint deflection. By combining information from the structures courses learnt so far (elementary structures and theory of structures) with programming, the project helps improve the understanding of truss behaviour. The results obtained show that the application can effectively analyse simple truss systems and provide a clear representation of their deformation

1.0 Introduction

Structural analysis is an important part of civil engineering because it helps engineers determine how structures behave under different loading conditions. Trusses are mostly used in structures like bridges, roof systems because they are strong, light and very economical. One of the standard methods used in analysing statically determinate trusses is the method of joints, which is based on the equilibrium of forces at each joint. Although the method is effective, manual calculations can become very difficult and time-consuming for bigger truss systems. This project aims to develop a Python-based application that simplifies the analysis process. The application allows users to the truss on a grid, apply loads and specify support reactions. The project combines programming and engineering concepts to create a simple and convenient application for truss analysis.

2.0 System Design

The application was developed using an object-oriented programming approach. The system is divided into several sections and classes that work together to perform structural analysis and graphical visualization.

The program consists of the following major components:

- Joint Class

The Joint class stores information related to each joint in the truss structure. The class contains:

- Joint identification (ID)
- X-coordinate
- Y-coordinate
- Applied load components
- Support conditions
- Reaction values
- Joint deflections

The class also converts applied load magnitude and angle into horizontal and vertical force components using trigonometric relationships.

- Member Class

The Member class represents each truss member connecting two joints. It stores:

- Member ID
- Start joint
- End joint
- Cross-sectional area
- Young's Modulus (E)
- Member length

- Internal member force
- Member condition (tension or compression)

The class also contains a method for calculating member length using coordinate geometry.

3.0 Graphical User Interface (GUI)

The graphical user interface was developed using Tkinter. The GUI allows users to:

- Define the number of joints and members
- Enter joint coordinates
- Specify support conditions
- Apply loads
- Define material properties
- Solve the truss system
- Display structural results

The interface includes buttons, labels, entry fields, combo boxes, message boxes, and result windows to improve user interaction.

- Solver Section:

The solver section performs the structural analysis using:

- Static equilibrium equations
- Method of Joints
- Matrix operations using NumPy
- Virtual Work Method for deflections

This section handles all numerical calculations within the application.

- Diagram Section

The diagram section uses Tkinter Canvas to visualize:

- Original truss geometry
- Deformed truss shape
- Support symbols
- Joint positions
- Member force labels
- Tension and compression members

4.0 Methodology

The methodology used in this project is based on principles of structural analysis and numerical computation.

4.1 Data Input

The user begins by defining:

- Number of joints
- Number of members
- Joint coordinates
- External loads
- Support conditions
- Material properties

These inputs are stored using the Joint and Member classes.

4.2 Determinacy Check

Before solving the truss, the application checks whether the structure is statically determinate using the equation:

$$m + r = 2j$$

Where:

- m = Number of members
- r = Number of reactions
- j = Number of joints

If the structure is unstable or indeterminate, the program displays an error message.

4.3 Support Reaction Calculation

The support reactions are calculated using static equilibrium equations:

- Sum of horizontal forces equals zero
- Sum of vertical forces equals zero
- Sum of moments equals zero

The `findReactions()` function performs these calculations automatically.

4.4 Method of Joints

The application uses the Method of Joints to determine unknown member forces. At each joint:

- Horizontal equilibrium is applied
- Vertical equilibrium is applied

NumPy is used to solve simultaneous equations for joints with two unknown member forces.

The computed member forces are stored and classified as:

- Tension
- Compression

4.5 Deflection Calculation

The Virtual Work Method (Unit Load Method) is used to compute joint deflections. A unit virtual load is applied at each joint in both x and y directions. Virtual member forces are then calculated, and deflections are determined using:

$$\Delta = \sum \frac{FfL}{EA}$$

Where:

- F = Real member force
- f = Virtual member force
- L = Member length
- E = Young's Modulus
- A = Cross-sectional area

4.6 Result Display and Visualization

After analysis, the application displays:

- Member forces
- Support reactions
- Joint deflections
- Equilibrium checks

The software also generates a graphical truss diagram showing:

- Original structure
- Deformed shape
- Tension members
- Compression members

5.0 Features of the Application

The developed software includes the following features:

- Addition of multiple joints and members.
- Input of joint coordinates and support conditions.
- Application of inclined external loads.
- Automatic conversion of loads into horizontal and vertical components.
- Automatic calculation of support reactions.
- Determinacy checking of truss structures.
- Member force calculation using the Method of Joints.
- Identification of tension and compression members.
- Deflection analysis using the Virtual Work Method.
- Graphical visualization of original and deformed truss structures.
- Interactive graphical user interface.
- Organized result display window.
- Equilibrium verification checks.
- Error handling for invalid structural configurations.

6.0 Tools and Technologies Used

6.1 Python Programming Language

Python was used because of its simplicity, readability, and strong support for engineering and scientific applications. It also supports object-oriented programming, which helped organize the application into different classes and functions.

6.1.1. Tkinter

Tkinter was used to create the graphical user interface (GUI). It provided:

- Buttons
- Labels
- Entry widgets
- Combo boxes
- Message boxes
- Canvas for graphical drawings

Tkinter made the application interactive and user-friendly.

6.1.2. NumPy

NumPy was used for:

- Matrix operations
- Numerical calculations
- Solving systems of equations
- Trigonometric computations

7.0 Benefits of the System

The developed application provides several advantages:

- Simplifies truss analysis by automating calculations.
- Reduces errors associated with manual computation.
- Saves time during structural analysis.
- Provides accurate numerical solutions.
- Improves visualization of structural behaviour.
- Helps students understand the Method of Joints and Virtual
- Combines structural engineering concepts with computer programming techniques.

8.0 Limitations

Despite its advantages, the current version of the software has some limitations:

- The application only analyses two-dimensional truss systems.
 - The program is limited to statically determinate trusses.
 - Dynamic loading conditions are not included.
 - Nonlinear material behaviour is not considered.
 - Temperature effects and support settlements are not included.
 - The software depends on correct user input for accurate results.
- Large and highly complex truss systems may increase computational time.

9.0 Conclusion

This project successfully demonstrates the development of a graphical Truss Solver Application using Python programming language. The software combines numerical computation with graphical visualization to create an efficient and user-friendly structural analysis tool. The application uses the Method of Joints to compute internal member forces and the Virtual Work Method to determine joint deflections. The integration of Tkinter and NumPy allowed the development of an interactive GUI capable of performing structural analysis accurately and efficiently. The software is particularly useful for civil engineering students and structural analysis learners because it simplifies complex calculations while improving understanding through graphical representation. The project also demonstrates how programming techniques can be applied to solve real engineering problems effectively.

